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$\times \log 0.0367$

$\Delta G_2^0 = +12312$ J

$\Delta G_2^0 = +12.312$ kJ

67.

Solution:

Given :

T = 33K

$N_2O_4 \rightleftharpoons 2NO_2$

Initial concentration 100% 0

Concentration dissociated 50% -

Concentration remaining at equilibrium 50% 100%

$K_{eq} = \frac{100}{50} = 2$

$\Delta G^0 = -2.303\ RT\log K_{eq}$

$\Delta G^0 = -2.303 \times 8.314 \times 33 \times \log 2$

$\Delta G^0 = -190.18$ J mol^{-1}

68.

Solution :

Given :

$\Delta H_f^0\ (SO_2) = -297$ kJ mol^{-1}

$\Delta H_f^0\ (SO_3) = -396$ kJ mol^{-1}

$SO_2 + \frac{1}{2} O_2 \rightarrow SO_3 \quad \Delta H_r^0 = ?$

$\Delta H_r^0 = (\Delta H_f^0)_{compound} - \Sigma (\Delta H_f)_{elements}$

$\Delta H_r^0 = \Delta H_f^0 (SO_3) - \left(\Delta H_f^0 (SO_2) + \frac{1}{2} \Delta H_f^0 (O_2) \right)$

$\Delta H_r^0 = -396$ kJ mol^{-1} $- (-297$ kJ mol$^{-1} + 0)$

$\Delta H_r^0 = -396$ kJ mol$^{-1} + 297$

$\Delta H_r^0 = -99$ kJ mol^{-1}

69.

Solution:

Given :

T = 298 K

$\Delta H = 400$ J mol$^{-1} = 400$ J mol^{-1}

$\Delta S = 0.2$ J K^{-1} mol^{-1}

$\Delta G = \Delta H - T\Delta S$

if T = 2000 K

$\Delta G = 400 - (0.2 \times 2000) = 0$

if T > 2000 K

ΔG will be negative.

The reaction would be spontaneous only beyond 2000K

70.

Solution :

Given :

T = 298K

$\Delta G_r^0 = -13.6$ kJ mol^{-1}

$= -13600$ J mol^{-1}

$\Delta G^0 = -2.303\ RT \log K_{eq}$

$\log K_{eq} = \frac{-\Delta G^0}{2.303\ RT}$

$\log K_{eq} = \frac{13.6 \text{kJ mol}^{-1}}{2.303 \times 8.314 \times 10^{-3} \text{JK}^{-1}\ \text{mol}^{-1} \times 298\ \text{K}}$

$\log K_{eq} = 2.38$

$K_{eq} = \text{antilog}\ (2.38)$

$K_{eq} = 239.88$